

Sympathetic Motor Neuron Dysfunction is a Missing Link in Age-Associated Sympathetic Overactivity

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Abstract

Overactivity of the sympathetic nervous system is a hallmark of aging. The cellular mechanisms behind this overactivity remain poorly understood, with most attention paid to likely central nervous system components. In this work, we hypothesized that aging also affects the function of motor neurons in the peripheral sympathetic ganglia. To test this hypothesis, we compared the electrophysiological responses and ion-channel activity of neurons isolated from the superior cervical ganglia of young (12 weeks), middle-aged (64 weeks), and old (115 weeks) mice. These approaches showed that aging does impact the intrinsic properties of sympathetic motor neurons, increasing spontaneous and evoked firing responses. A reduction of KCNQ channel currents emerged as a major contributor to age-related hyperexcitability. Thus, it is essential to consider the effect of aging on motor components of the sympathetic reflex as a crucial part of the mechanism involved in sympathetic overactivity.

eLife assessment

This study presents **valuable** observations indicating that the excitability of cultured sympathetic motor neurons increases in neurons cultured from aged mice, and is inversely correlated with the amplitude of KCNQ currents. The alterations in membrane excitability are relevant for aging-related changes in neuronal membrane properties. While the study documents interesting changes in membrane excitability in cultured neurons with aging, the mechanisms underlying these changes are not clear and physiological relevance of the results to the intact circuits is **incomplete**.

Introduction

This study focuses on the mechanisms underlying the deterioration of the sympathetic nervous system as we age. With advancing age, the sympathetic nervous system tends to become overactive, evidenced by increased electrical activity in sympathetic nerves [Hart et al. \(2009\)](#) ; [Ito et al. \(1986\)](#) ; [Narkiewicz et al. \(2005\)](#) ; [Wallin et al. \(1974\)](#) . This overactivity, observed in

humans and animal models, leads to heightened release of norepinephrine over organs [Esler et al. \(1995\)](#); [Goldstein et al. \(1983a,b\)](#); [Veith et al. \(1986\)](#); [Ziegler et al. \(1976\)](#), triggering compensatory processes and cellular deterioration [Hogikyan and Supiano \(1994\)](#). Understanding the physiology and pathophysiology of this age-associated overactivity is crucial, as many common age-driven diseases are linked to sympathetic overactivity. For instance, dysregulation of norepinephrine release is associated with age-related hypertension and arrhythmias. However, the neuronal mechanisms underlying the overactivity of the sympathetic nervous system remain to be elucidated.

During execution of sympathetic reflexes, three essential cellular components of the nervous system come into play: sensory neurons for detecting and transmitting signals of internal conditions, neurons of the sympathetic nuclei in the brain for integrating information and responding to internal changes, and the sympathetic motor neurons in the ganglia for receiving and delivering information to target organs (**Figure 1**). Age-associated overactivity of the sympathetic nervous system has been predominantly ascribed to changes in the central system, specifically focusing on the hypothalamic and brainstem nuclei. In the paraventricular nucleus of the hypothalamus, the activation of glutamatergic neurons through leptin pathways has emerged as a significant contributor to sympathetic overactivity [Shi et al. \(2015, 2020\)](#). At the same time, glial senescence and subsequent inflammatory mechanisms in brainstem nuclei also have been linked to sympathetic nervous system overactivity during aging [Balasubramanian et al. \(2021\)](#). However, one key question remains: Is age-related sympathetic overactivity limited solely to changes in the central system?

Aging impacts the morphology and calcium responses of sympathetic motor neurons, the peripheral component of the sympathetic reflex. Notably, sympathetic innervation to the spleen and lymph nodes reduces, while thymus innervation increases with age [Madden et al. \(1997\)](#). Similarly, innervation to vascular smooth muscle changes with age, and the extent and implications of these changes depend on the specific target and neuropeptide content [Andrews et al. \(1996\)](#); [Andrews and Cowen \(1994\)](#); [Burnstock \(1990\)](#). Additionally, aging affects the function of sympathetic motor neurons, as evidenced by altered calcium responses. Old sympathetic motor neurons display reduced expression of SERCA pumps, an essential component for calcium transport into the endoplasmic reticulum [Buchholz et al. \(2007\)](#); [Pottorf et al. \(2000\)](#). The results of these experiments suggest that aging has a distinct influence on the intrinsic function of sympathetic motor neurons, irrespective of any changes in the central system. As a result, the main objective of our study was to examine directly the impact of aging on the intrinsic membrane electrical properties of sympathetic motor neurons (**Figure 1**).

Our work was structured around three main objectives: 1) To standardize the isolation of sympathetic motor neurons from adult to old age in mice, 2) To compare the spontaneous and evoked electrical activity of sympathetic motor neurons at three life stages, 3) To identify potential molecular candidates underlying the observed altered electric properties and neuronal activity.

Results

Sympathetic motor neurons from old mice are healthy in culture

Our first goal was to assess the viability of neurons isolated from young adult and old mice. We isolated sympathetic motor neurons enzymatically from the left and right superior cervical ganglia (SCG) of young adults (12 weeks old) and old animals (115 weeks old, **Figure 2A**). The same enzymatic digestion was used for both ages. Single neurons exhibited no visible neurite growth before 18 hours (**Figure 2B**, left); however, after 72 hours in culture, both young and old sympathetic motor neurons showed evident neurite growth (**Figure 2B**, right). The mean diameter of the cell soma was found to be similar in young ($19.7 \pm 0.6 \mu\text{m}$) and old ($19.6 \pm 0.6 \mu\text{m}$)

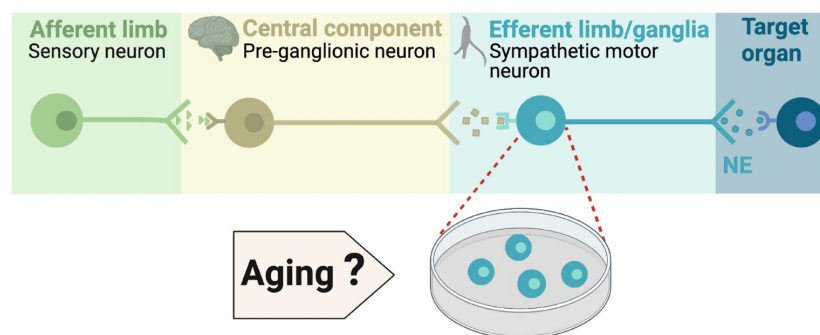


Figure 1.

Components of the sympathetic reflex.

Schematic of the components of the sympathetic autonomic reflex: Sensory neurons send information to the central component (pre-ganglionic neurons), where information is processed. Then, sympathetic motor neurons receive information from pre-ganglionic neurons to transmit it to the target organ. This research was focused on evaluating the age-related changes in the function of sympathetic motor neurons.

neurons after 24 and 72 hours in culture (**Figure 2C**). The dendritic arborization after 24 hours in culture was comparable between young ($37 \pm 2 \mu\text{m}^2$) and old ($42 \pm 3 \mu\text{m}^2$) neurons. However, after 72 hours in culture, the dendritic arborization in old neurons ($105 \pm 7 \mu\text{m}^2$) was more extensive compared to that of young neurons ($74 \pm 5 \mu\text{m}^2$) (**Figure 2D**). We continued the culture for seven days and noted that the neurites grew until they contacted other neurons in the same dish. Furthermore, glial cells became apparent at this stage in cultures of young and old cells. The ability of neurons to regenerate neurites and contact each other was considered an indicator of viability and health. We ruled out the possibility that any functional differences between neurons from animals of different ages were due to potential damage to old neurons during the isolation and culture process. From now on, we report electrical measurements on neurons that had been incubated for approximately just 12–18 h after isolation.

Old sympathetic motor neurons fire action potentials spontaneously

To investigate whether aging alters the function of sympathetic motor neurons, we first compared the spontaneous activity and passive electrical properties of neurons isolated from mice at different ages: 12 weeks (young), 64 weeks (middle age), and 115 weeks (old). For reference, we also provide comparable human ages (**Figure 3A**). Spontaneous activity was recorded using the current clamp modality without applying any holding or current stimulus. Electrical access to the cell was obtained using the perforated patch technique (See Methods for details). Young neurons had a resting membrane potential (RMP) of $-64 \pm 1 \text{ mV}$ and very little spontaneous activity, which aligns with the expected behavior of motor neurons that respond to presynaptic commands (**Figure 3B-C**). Only 3% exhibited spontaneous firing (**Figure 3D**). In contrast, middle-aged neurons showed an RMP of $-58 \pm 1 \text{ mV}$, and 37% of these neurons displayed spontaneous firing (**Figure 3B-D**). The depolarized RMP and the increased number of neurons spontaneously firing persisted in old neurons, where the RMP was $-54 \pm 1 \text{ mV}$ and 58% of neurons exhibited spontaneous firing (**Figure 3B-D**). Despite the higher percentage of old neurons firing spontaneously compared to middle-aged neurons, the mean firing frequency in old neurons (60 ± 20 action potentials (APs)/minute) was significantly lower than that observed in middle-aged neurons (177 ± 41 APs/minute; **Figure 3E**).

We measured the input resistance, another passive membrane property, in cells polarized to a potential of -65 mV by injecting a negative holding current that did not exceed -50 pA . Cells requiring larger holding currents were excluded. From the response to additional negative current pulses (-40 to 0 pA , **Figure 3F**) we calculated the input resistance ($\Delta V/\text{injected current}$). At these membrane potentials, without activation of voltage-activated ion channels, the change in voltage is proportional to the injected current. Input resistance was not affected by age. Young adult neurons ($0.8 \pm 0.09 \text{ G}\Omega$), middle-aged neurons ($1.1 \pm 0.06 \text{ G}\Omega$), and old neurons ($1.1 \pm 0.11 \text{ G}\Omega$) exhibited similar input resistance (**Figure 3G**). In conclusion, our findings suggest that aging leads to a gradual depolarization of the RMP without significant alteration of the input resistance.

Sympathetic motor neurons from old mice are more responsive to electrical stimulation

During sympathetic reflexes, the peripheral motor neurons increase their firing frequency in response to commands from the central sympathetic nuclei. These commands can be mimicked as direct electrical stimulation, as shown in **Figure 4A**. To test whether aging increases the sensitivity of sympathetic motor neurons to electrical stimulation, neurons were injected with 1-second current steps of amplitude ranging from 10 pA to 100 pA starting from a potential of -65 mV (“ 0 pA current”) (**Figure 4B**). Young neurons exhibited a characteristic firing pattern demonstrated in the left panel of **Figure 4C**. With injections of 10 pA , no AP was elicited (red trace), and with a 40 pA stimulus lasting 1 second, only one AP was evoked (black trace),

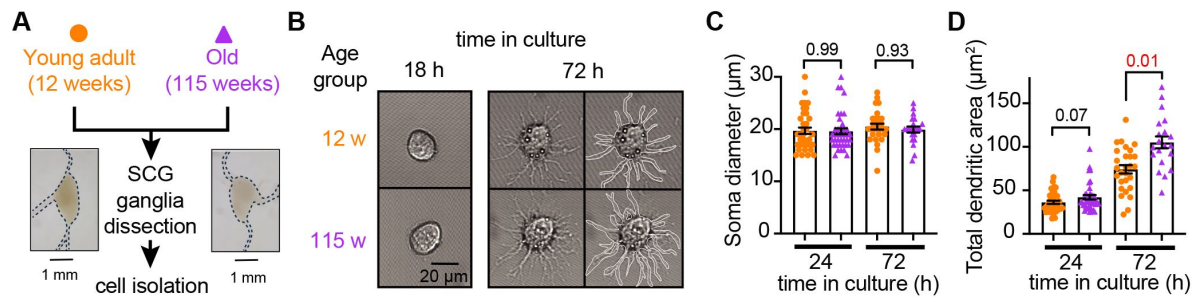


Figure 2.

Sympathetic motor neurons from old mice are healthy in culture.

(A) Diagram of the experimental approach: Sympathetic motor neurons were isolated from the superior cervical ganglia (SCG) of 12- and 115-week-old mice. The ganglia did not show morphological differences between ages. **(B)** Differential Interference Contrast images of sympathetic motor neurons in culture for 18 h and 72 h. Right images at 72 h outline the dendritic area measured. w, weeks. **(C)** Comparison of soma diameter between neurons isolated from 12 or 115 weeks of age at two-time points in culture. Orange circles showed single cells from 12-week-old mice while purple triangles show single cells from 115-week-old mice. **(D)** Comparison of the total dendritic area as a proxy for neurite regeneration in culture. Data points are from N = 3 animals, n = 38 cells, from 12 weeks old, and N = 3 animals, n = 39 cells, from 115 weeks old. p-values are shown at the top of the graphs. Red values indicate p-values < 0.05 while black values indicate p-values > 0.05.

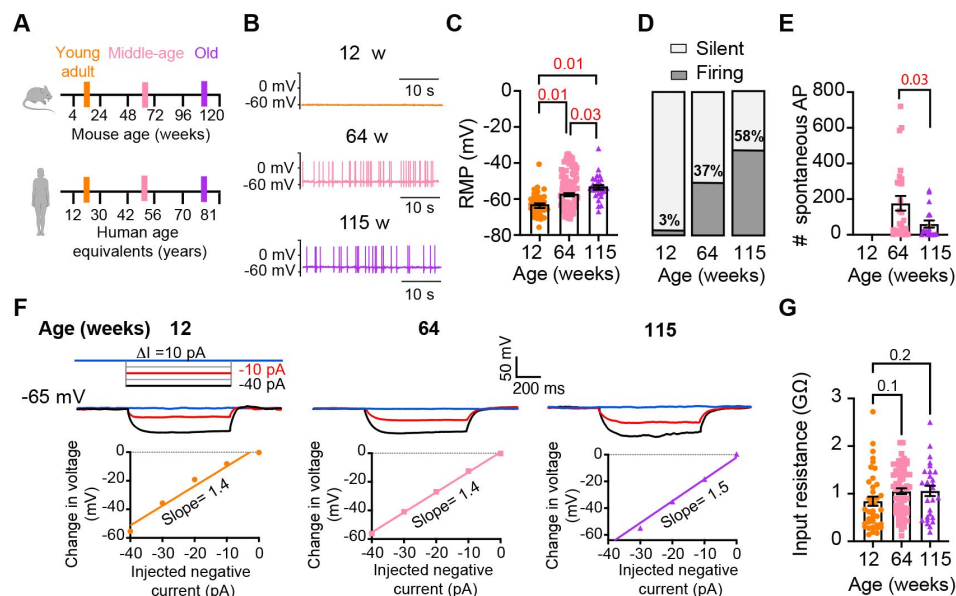


Figure 3.

Sympathetic motor neurons from old mice fire action potentials spontaneously.

(A) Schematic of ages in mice, and equivalent in humans, that were used to compare the functional responses of sympathetic motor neurons. **(B)** Representative membrane potential recordings of spontaneous activity from neurons isolated from 12-, 64-, and 115-week-old mice. **(C)** Comparison of the Resting Membrane Potential (RMP) between different ages. **(D)** Comparison of percentage of silent and firing neurons between different ages. **(E)** Comparison of the number of action potentials (AP) fired spontaneously in one minute between different ages. **(F)** Top: Representative passive responses to hyperpolarizing stimuli from neurons isolated from 12-, 64-, and 115-week-old mice. Blue traces correspond to 0 pA injection, red traces to -10 pA, and black traces to -40 pA. Bottom: voltage-current relationship of top recordings. **(G)** Comparison of the input resistance between different ages. Data points are from N = 4 animals, n = 35 cells, from 12-week old, N = 6 animals, n = 65 cells, from 64-week old, and N = 4 animals, n = 32 cells, from 115-week old. p-values are shown at the top of the graphs. Red p-values indicate p-values < 0.05, while black p-values indicate p-values > 0.05.

illustrating the phenomenon known as spike adaptation that is typical of young sympathetic motor neurons. In contrast, both middle-aged and old neurons displayed a different response. A 10 pA stimulus was sufficient to elicit APs in middle-aged and old neurons (**Figure 4D-E**), and in middle-aged neurons, the 40 pA stimulus induced the firing of eleven APs in a typical example (**Figure 4D**), and similarly, in old neurons, it led to the firing of ten APs (**Figure 4E**).

To assess the impact of age on the excitability of sympathetic motor neurons in a population sample, we compared stimulus-response curves, rheobase values (minimum current needed to elicit at least one AP), and the number of APs elicited with the maximum stimulus. Our results revealed that middle-aged and old neurons fired more APs with each stimulus (**Figure 4F**) than young neurons. Older neurons exhibited a reduced threshold for eliciting APs (64 weeks: 20 ± 2 pA; 115 weeks: 23 ± 5 pA) compared to young neurons (34 ± 4 pA, **Figure 4G**). Furthermore, older neurons fired more APs with a 100 pA current injection (64 weeks: 15 ± 1 APs; 115 weeks: 15 ± 2 APs) compared to young neurons (12 weeks: 7 ± 1 APs, **Figure 4H**). These findings support the concept that aging leads to increased excitability of sympathetic motor neurons.

Analysis of neuronal subpopulations

Sympathetic motor neurons display stereotyped distinct repetitive firing patterns, which classify them into three categories: tonic (Class I), phasic (Class II), and adapting (Class III) [Kim et al. \(2019\)](#); [Malin and Nerbonne \(2000, 2001\)](#); [Springer et al. \(2015\)](#). In our experiments, we classified cells based on their responses to a supra-threshold current injection (20 pA above their rheobase), ensuring that the firing response was not saturated. Representative traces of firing patterns from middle-aged neurons are illustrated in **Figure 5A**. The increased number of APs at maximal stimulus compared with 20 pA above the rheobase, shows that the firing response was not saturated during classification (**Figure 5B**). In agreement with previous reports, subpopulations showed differences in frequency-stimulus curves (**Figure 5C**). Also, tonic cells have a more depolarized RMP (**Figure 5D** left, tonic = -53 ± 2 mV, phasic = -61 ± 1 mV, adapting = -60 ± 2 mV), while adapting neurons have a lower input resistance (**Figure 5D** right, tonic = 1.13 ± 0.08 G Ω , Phasic = 1.16 ± 0.09 G Ω , Adapting = 0.55 ± 0.18 G Ω). Tonic cells were also more spontaneously active (**Figure 5D** middle). In general, the responses from the three neuronal subpopulations are consistent with previous reports.

Next, we examined the effect of aging on neuronal subpopulations. We observed an altered subtype distribution in older neurons (**Figure 5E**, p-value < 0.0001 using a contingency table and Chi-square/Fisher test). The percentage of adapting neurons decreased in middle age (10%) and old (12%) compared to young (29%). The percentage of phasic neurons decreased only in old (42%) compared to middle-aged (56%) and young (55%) neurons. Accordingly, the percentage of tonic neurons increased in middle-aged (34%) and old (46%) compared to young (19%) neurons. These findings highlight how aging impacts the distribution of firing subtypes in sympathetic motor neurons.

While age does impact the distribution of firing subtypes, the question remains whether intrinsic properties are affected within each subpopulation across different ages. Therefore, we analyzed the intrinsic properties across subpopulations between different age groups. The RMP of adapting and phasic firing neurons was more depolarized in middle age and old compared to young neurons (**Figure 5F**). The RMP of tonic neurons tended to become more depolarized with age, but the effect was not significant (**Figure 5F**). Similar to the observation when the entire population was analyzed, the input resistance was not significantly different with age within phasic and adapting neurons. The input resistance of tonic neurons was significantly larger only in old (1.44 ± 0.16 G Ω) neurons compared to young (0.93 ± 0.16 G Ω) neurons (**Figure 5G**). This analysis suggested that, regardless of the subpopulation type, older neurons tend to have more depolarized RMP with no changes in input resistance.

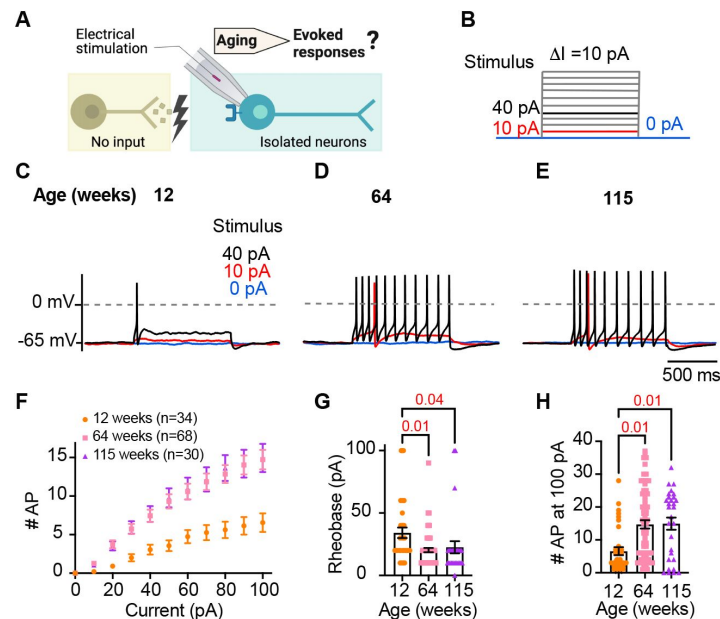


Figure 4.

Sympathetic motor neurons from old mice are more responsive to electrical stimulation.

(A) In the context of the sympathetic reflex, motor neurons increase their firing frequency in response to inputs from the sympathetic nuclei in the brain. In this study, motor neuron responses were measured using electrical stimulation at different ages. **(B)** Stimulation protocol to mimic preganglionic input. **(C-E)** Representative voltage responses of sympathetic motor neurons from 12-(C), 64-(D), and 115-week-old mice (E). Blue, red, and black traces are in response to 0, 10, and 40 pA current injection respectively. The dotted line shows 0 mV as reference. **(F)** Comparison of stimulation-response curves of the number of APs vs. injected current between different ages. **(G)** Comparison of the minimum current injected that elicited at least one AP (Rheobase) between different ages. **(H)** Comparison of the number of APs fired at the maximum stimulus (100 pA) between different ages. Data were collected from N = 4 animals, n = 34 cells, from 12 weeks old, N = 6 animals, n = 68 cells, from 64 weeks old, and N = 4 animals, n = 30 cells, from 115 weeks old. p-values are shown at the top of the graphs.

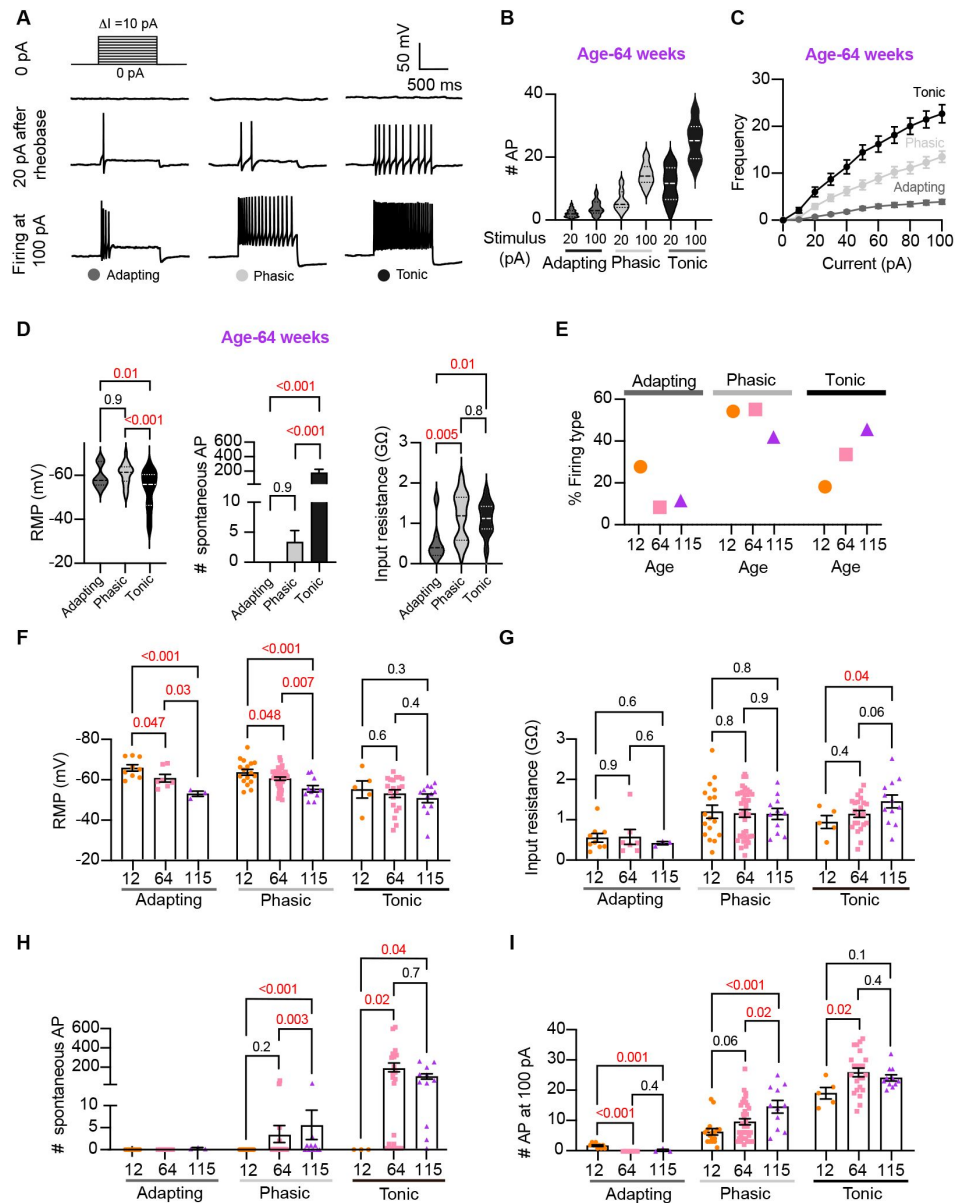


Figure 5.

Analysis of neuronal subpopulations. Aging shifts neuronal population toward tonic firing.

(A) Representative recordings from three different neurons illustrate the variability in the response to show the method used to classify cells by their firing pattern as adapting (left), phasic (center), or tonic (right). Responses to stimulation of 0 (top), 20 pA above the rheobase (middle), and 100 pA (bottom). The classification was based on the response 20 pA above the rheobase. (B) Comparison of the number of APs elicited by current injections of 20 pA more than rheobase or 100 pA between classes. Data are from 64-week-old mice. (C) Comparison of the stimulus-frequency curves between classes. Data are from 64-week-old mice. (D) Comparison of RMP, (left), number of spontaneous APs (middle), and input resistance (right) between classes. All data are from 64-week-old mice. (E) Comparison of the percentage of neuronal firing subtypes between different ages. (F) Comparison of the RMP between ages and divided into neuronal subpopulations. (G) Comparison of the input resistance between ages and divided into neuronal subpopulations. (H) Comparison of the number of spontaneous APs between ages and neuronal subpopulations. (I) Comparison of the number of APs in response to maximum stimulation (100 pA) between ages and neuronal subpopulations. Data points for the adapting subpopulation are 9 cells from 12-weeks-old, 7 from 64-weeks-old, and 3 from 115-weeks-old. Data points for the phasic subpopulation are 17 cells from 12-weeks-old, 38 from 64-weeks-old, and 11 cells from 115-weeks-old. Data points for the tonic subpopulation are 5 cells from 12-weeks-old, 23 from 64-weeks-old, and 11 cells from 115-weeks-old. Data points are also from a total of 4 12-week-old mice, 6 64-week-old mice, and 4 115-week old mice. Red values indicate p-values < 0.05, while black values indicate p-values > 0.05.

Interestingly, aging leads to an increase in the number of spontaneous APs in phasic and tonic neurons (**Figure 5H**). In the phasic class, middle-aged neurons fired 3 ± 2 APs per minute and old neurons fired 5 ± 3 APs, compared to zero APs in phasic young neurons. In the tonic class, middle-aged neurons fired 179 ± 46 APs and old neurons fired 86 ± 27 APs, in striking contrast to zero APs in tonic young neurons. Next, we compared the maximum number of APs elicited by the strongest stimulus (100 pA, **Figure 5I**). In phasic neurons, the number of APs at 100 pA increased by 53% in middle age (9.5 ± 1.0 APs) and 133% in old (14.5 ± 2.1 APs) compared to young neurons (6.2 ± 1.0 APs). In tonic firing neurons, the number of APs at 100 pA increased by 36% in middle age (25 ± 1.4 APs) and 27% in old (24.1 ± 1.0 AP) compared to young neurons (19.0 ± 1.9 APs).

We conclude that the changes observed when categorizing neuronal classes occur across the entire population. In addition, aging is associated with a shift in the proportion of neurons falling in each class. As a result, we hypothesized that the underlying age-related molecular mechanism is broadly shared among sympathetic motor neurons and plays a role in controlling the firing frequency.

Aging reduces KCNQ current

We next directed our attention to identifying molecular candidates underlying changes in membrane excitability. Sympathetic motor neurons express at least one voltage-gated sodium channel isoform ($\text{Na}_v1.7$) [Schofield et al. \(2008\)](#) and [Toledo-Aral et al. \(1997\)](#) and several voltage-gated potassium channels [Dixon and McKinnon \(1996\)](#); [Shi et al. \(1997\)](#). Specifically, K_v4 , K_v2 , and K_v7 (KCNQ) channels are crucial in controlling the RMP, rheobase, firing frequency, and spike adaptation [Liu and Bean \(2014\)](#); [Malin and Nerbonne \(2000, 2001\)](#). These channels are of particular interest as potential contributors to the age-related alterations in neuronal excitability that we observed.

Previous work by our group and others demonstrated that cholinergic stimulation (with oxotremorine M, Oxo-M) leads to a decrease in KCNQ currents and increases the excitability of sympathetic motor neurons at young ages. Thus, **Figure 6A** shows a voltage response (measured in current-clamp mode) and a consecutive KCNQ current recording (measured in voltage-clamp mode) in the same neuron upon stimulation of cholinergic type 1 muscarinic receptors. It illustrates the temporal correlation between the decrease of KCNQ current with the increase in excitability and firing of APs [Kruse and Whitten \(2021\)](#). This strong dependence led us to hypothesize that, like Oxo-M, aging decreases KCNQ current, leading to a depolarized RMP and hyperexcitability (**Figure 6B**). For these experiments, we measured the RMP and evoked activity using perforated patch, followed by the amplitude of KCNQ current using a whole-cell voltage clamp in the same cell. We also measured the membrane capacitance as a proxy for cell size. Interestingly, KCNQ current density was smaller by 29% in middle age (7.5 ± 0.7 pA/pF) and by 55% in old (4.8 ± 0.7 pA/pF) compared to young (10.6 ± 1.5 pA/pF) neurons (**Figure 6C-D**). The average capacitance was similar in young (30.8 ± 2.2 pF), middle-aged (27.4 ± 1.2 pF), and old (28.8 ± 2.3 pF) neurons (**Figure 6E**), suggesting that aging is not associated with changes in cell size of sympathetic motor neurons, and supporting the hypothesis that aging alters the levels of KCNQ current. Interestingly, we observed that KCNQ2 protein levels were 1.5 ± 0.1 -fold higher in old compared to young neurons (**Figure 6F-G**). The decrease in KCNQ current and the increase in the abundance of KCNQ2 protein suggest a potential compensatory mechanism that occurs during aging, which we are actively investigating in an independent study. To explore the hypothesis that a reduction in KCNQ current is responsible for the age-associated depolarization of the RMP, we compared these two parameters measured in the same cells. We observed a correlation between KCNQ current and RMP in young (coefficient of determination (r^2) = 0.22, p-value for the correlation fit = 0.007, **Figure 6H** and **6K**) and middle-aged neurons (r^2 = 0.20, p-value for the correlation fit = 0.002, **Figure 6I** and **6K**). In old neurons, the KCNQ current and RMP were no longer correlated (r^2 = 0.05, p-value for the correlation fit = 0.1, **Figure 6J** and **K**). **Figure 6K** shows the decrease in the coefficient of determination (r^2) with aging. Similarly, the KCNQ current

amplitude also correlated well with the number of APs elicited at 100 pA in young and middle-aged neurons but not in old ones. The variance in KCNQ current amplitude explained 32% of the variation in the number of APs in young ($r^2 = 0.32$, p-value for the correlation fit = 0.001, **Figure 6L**) and 24% in middle-aged neurons ($r^2 = 0.24$, p-value for the correlation fit = 0.0001, **Figure 6M**). In old neurons, the variance in KCNQ current amplitude explained only 0.05% of the variation in the number of APs ($r^2 = 0.05$, p-value for the correlation fit = 0.15, **Figure 6N**). **Figure 6O** shows the decrease in the coefficient of determination with aging for the number of APs. These analyses support the hypothesis that a reduction in KCNQ current alters the electrical properties, and in the case of old neurons, the marked decrease in KCNQ current compromises its role in maintaining the RMP and spontaneous firing.

Other voltage-gated sodium and potassium currents are not altered with aging

Loss of voltage-gated potassium channel function, including the current of K_{V2} and K_{V4} , has also been invoked in aging and age-associated memory decline [Fenyves et al. \(2021\)](#); [Frazzini et al. \(2016\)](#); [Navarro-Garcia et al. \(2022\)](#); [Sesti \(2016\)](#); [Simkin et al. \(2015\)](#); [Yu et al. \(2019\)](#). Therefore, we also looked for potential age-associated changes in other voltage-gated potassium currents. We used a recording solution designed to abolish sodium, calcium, and potassium currents mediated by both calcium-activated potassium channels (BK channels) and KCNQ channels. This recording solution contained 100 nM tetrodotoxin (TTX) to suppress voltage-gated sodium channels, 100 μ M Cd^{2+} to inhibit calcium channels and 10 μ M XE-991 to block KCNQ channels. Next, we compared the outward current density before and after application of a cocktail of 100 nM phiroxotoxin and 100 nM guangxitoxin to block K_{V4} and K_{V2} channels. **Figure 7A** and **C** show representative traces and the comparison of the current density between young and old neurons. We did not observe significant differences in the potassium current insensitive to XE-991 between groups (young = 363 ± 20 pA/pF, old = 342 ± 30 pA/pF, **Figure 7A** and **B**) nor in the K_{V2} and K_{V4} sensitive currents (young = 157 ± 31 pA/pF, old = 110 ± 17 pA/pF, **Figure 7C** and **D**). In nociceptive neurons, similar to sympathetic neurons, modulation of $Na_v1.7$ channels has been linked to increased excitability [Akin et al. \(2021\)](#); [Dib-Hajj et al. \(2013\)](#); [Li et al. \(2018\)](#). Additionally, exogenous expression of $Na_v1.7$ channels in non-excitatory cells induces cell senescence [Warnier et al. \(2018\)](#), further highlighting a significant connection between aging and voltage-gated sodium channels. To investigate the hypothesis that aging might contribute to hyperexcitability through alterations in sodium currents in sympathetic motor neurons, we conducted experiments using 5 ms voltage steps in a low-sodium recording solution. We found that the sodium current density was similar in young (58.9 ± 5.4 pA/pF), middle-aged (59.0 ± 5.3 pA/pF), and old (50.1 ± 5.7 pA/pF) neurons (**Figure 7E-G**). In conclusion, our data indicate that aging does not alter the Na_v and K_v currents insensitive to XE-991 in sympathetic motor neurons.

Pharmacological inhibition of KCNQ channels mimics aged phenotype while pharmacological activation of KCNQ channels mimics young phenotype

To further test the hypothesis that the decrease of KCNQ current is responsible for the hyperexcited state in old neurons, we used a pharmacological approach targeting KCNQ channels and assessed whether the age-dependent phenotype could be mimicked or reversed. To inhibit KCNQ channels, we used linopirdine at a concentration of 25 μ M. Linopirdine inhibits both KCNQ2 and KCNQ3 channels with an IC_{50} of around 5 μ M [Attali et al. \(2023\)](#) and with an inhibition time constant of around 3 s at negative potentials [Greene et al. \(2017\)](#). In the presence of linopirdine,

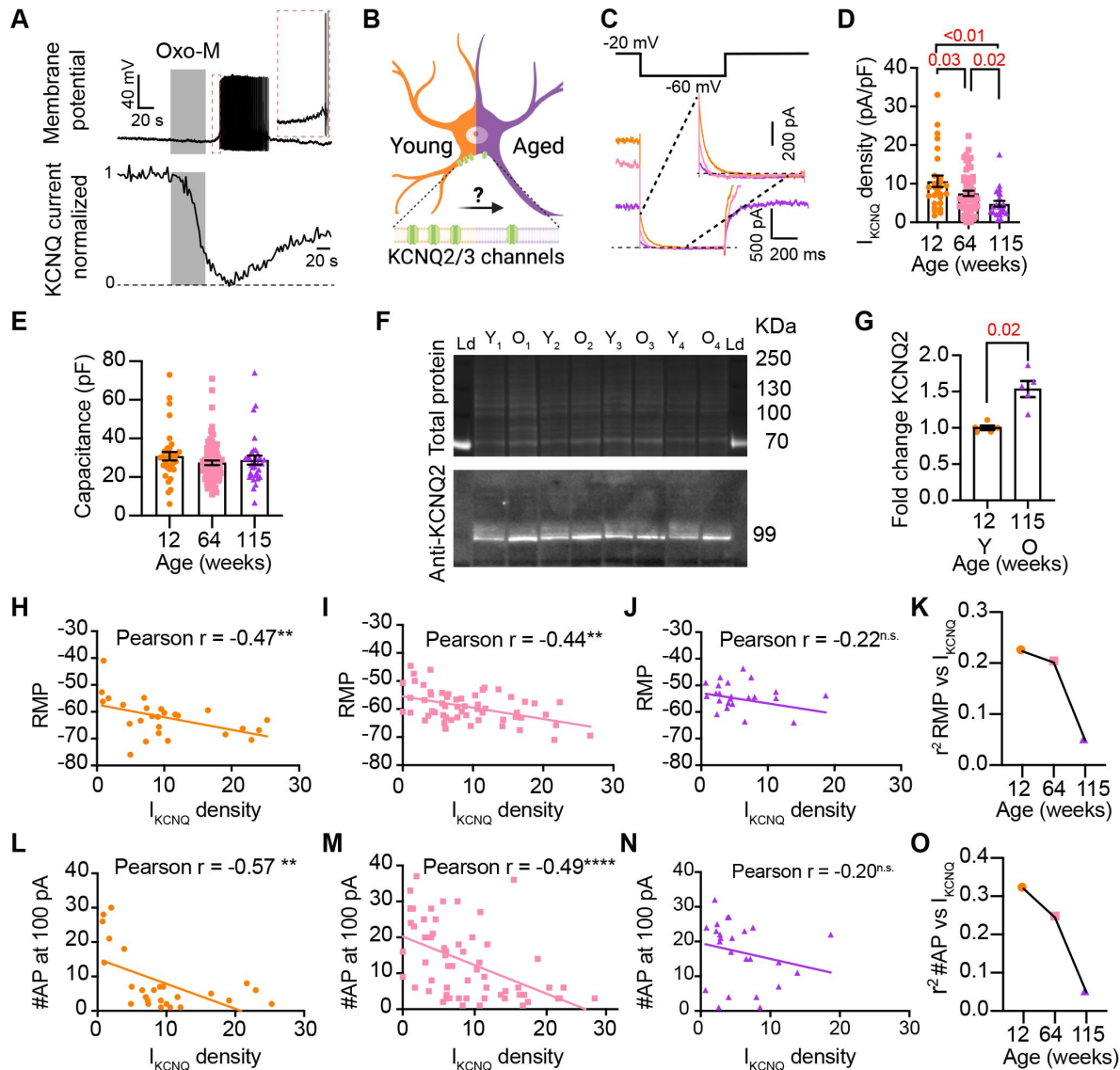


Figure 6.

Sympathetic motor neurons from old mice show a KCNQ current reduction. **(A)** Recordings illustrating the relevance of KCNQ current for controlling membrane potential and firing in sympathetic motor neurons. Top: voltage response to inhibition of KCNQ channels with 10 μ M Oxo-M, bottom: normalized KCNQ current recording in response to oxo-M in the same cell after going whole cell. **(B)** Schematic representation of the hypothesis that aged cells show reduced activity of KCNQ channels. **(C)** KCNQ current recordings from neurons isolated from mice of different ages (orange 12 weeks old, pink 64 weeks old, and purple 115 weeks old) in response to a voltage step (top). The inset shows an expanded view of the current tail. **(D)** Comparison of KCNQ current density between different ages. Data points are from N = 5 animals, n = 27 cells, from 12 weeks old, N = 8 animals, n = 62 cells, from 64 weeks old, and N = 5 animals, n = 24 cells, from 115 weeks old. **(E)** Comparison of capacitance between ages. **(F)** blot stained for total and KCNQ2 protein collected from superior cervical ganglia from 12- and 115-weeks-old mice. **(G)** Comparison of fold change of KCNQ2 abundance relative to total protein between ages (n = 5 blots from different mice). **(H-J)** Linear correlation between RMP and KCNQ current density in 12-weeks old **(H)**, 64-weeks old **(I)**, and 115-weeks old mice **(J)**. **(K)** Comparison of the determination coefficient for the RMP and KCNQ current between ages. **(L-N)** Linear correlation between the number of APs at 100 pA and KCNQ current density in 12-week-old **(L)**, 64-week-old **(M)**, and 115-week-old mice **(N)**. **(O)** Comparison of the determination coefficient for the maximum #AP and KCNQ current between ages. Data points are from N = 4 animals, n = 26 cells, from 12 weeks old, N = 6 animals, n = 58 cells, from 64 weeks old, and N = 4 animals, n = 24 cells, from 115 weeks old.

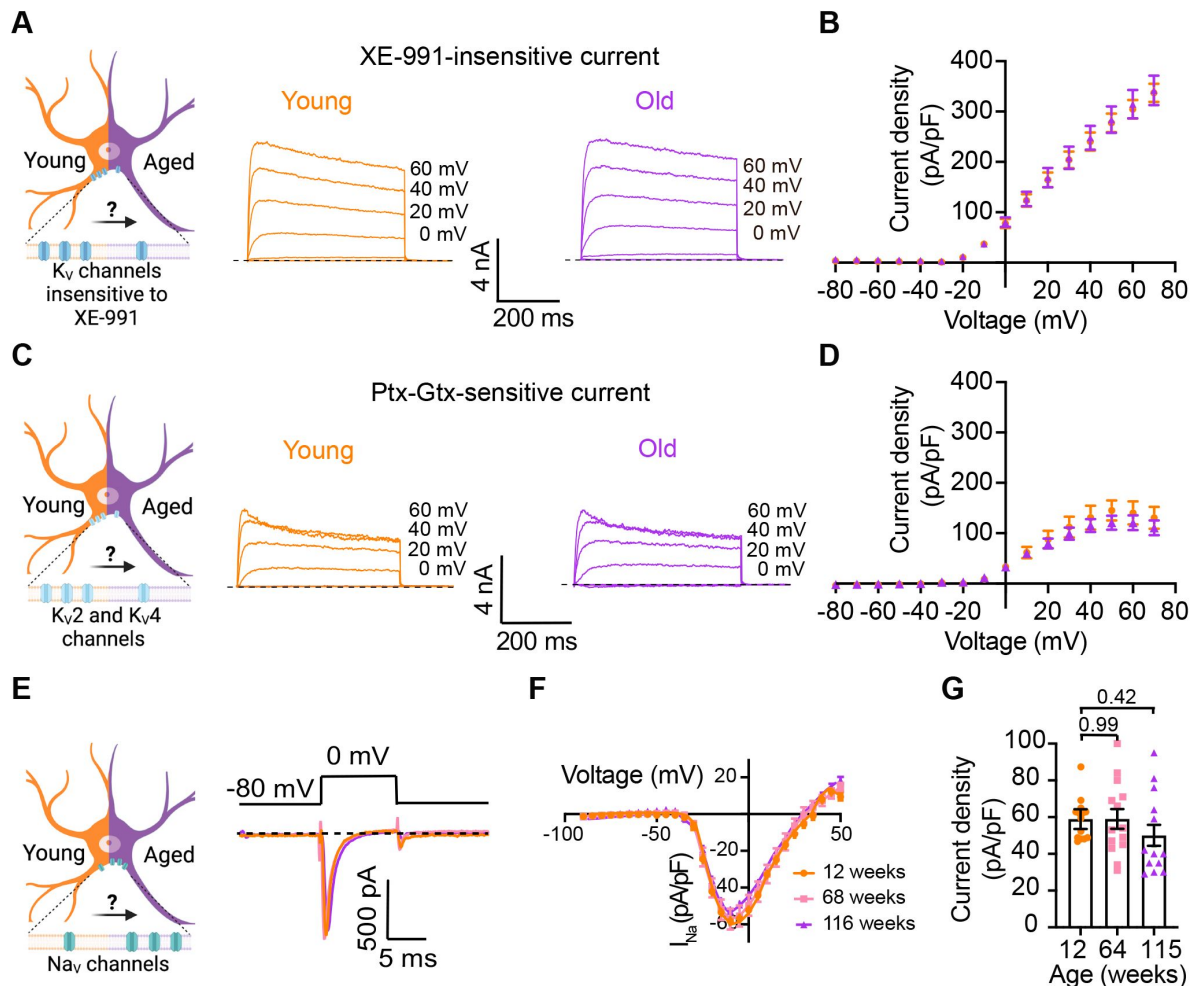


Figure 7.

Sympathetic motor neurons from old mice did not show changes in sodium currents or XE-991-insensitive potassium currents.

(A) Left: Schematic of the hypothesis that K_V channels insensitive to XE-991 are reduced in aged cells. Right: Family of outward currents in the presence of KCNQ channel blocker, 100 nM XE-991, in 12-week-old, and 115-week-old mice. Text at the right of each current trace corresponds to the voltage used to elicit that current. **(B)** Current-voltage relationship of the steady-state K_V currents insensitive to XE-991 from cells isolated from 12-week-old and 115-week-old mice. Data points of K_V currents are from N = 3 animals, n = 9 cells, from 12-weeks old, and N = 3 animals, n = 8 cells, from 115-weeks old. **(C)** Left: Schematic of the hypothesis that K_{V2} (Guangxitoxin-1E sensitive, Gtx) and K_{V4} (Phrixotoxin-1 sensitive, Ptx) channels are reduced in aged cells. Right: Family of Ptx- and Gtx-sensitive outward currents in 12-week-old and 115-week-old mice. Text at the right of each current trace corresponds to the voltage used to elicit that current. **(D)** Current-voltage relationship of the steady-state K_V currents sensitive to Ptx- and Gtx from cells isolated from 12-week-old and 115-week-old mice. Data points of K_{V2} and K_{V4} currents are from N = 3 animals, n = 12 cells, from 12-weeks old, and N = 3 animals, n = 8 cells, from 115-weeks old. **(E)** Left: Schematic of the hypothesis that Na_V channels sensitive to TTX are increased in aged cells. Right: Representative sodium current recordings from neurons isolated from mice of different ages (orange 12-week-old, pink 64-week-old, and purple 115-week-old) in response to a voltage step (top). **(F)** Current-voltage relationship of the peak sodium current for different ages. **(G)** Comparison of sodium current density between different ages. Data points of Na_V currents are from N = 3 animals, n = 12 cells, from 12-weeks old, N = 3 animals, n = 14 cells, from 64-weeks old, and N = 3 animals, n = 13 cells, from 115-weeks old. p-values are shown at the top of the graphs.

the resting membrane potential of young neurons became less negative, from -64.3 ± 5.1 mV to -50.4 ± 5.1 mV on average. In 5 out of 8 cases, neurons that were not firing started firing after washing linopirdine into the bath (**Figure 8A** [↗](#) and **B** [↗](#)).

Conversely, we used retigabine at a concentration of 10 μ M to activate KCNQ channels. Retigabine activates KCNQ2 with an IC_{50} of 2.5 μ M and KCNQ3 with an IC_{50} of 600 nM [Attali et al. \(2023\)](#) [↗](#). The onset of its action takes about 3 minutes, according to information provided by the manufacturer. In the presence of retigabine, the resting membrane potential of old neurons became more negative, from -49.9 ± 3.8 mV to -74.7 ± 3.0 mV on average. Aged neurons that were firing action potentials stopped firing after the activation of KCNQ channels, and some of these fired again after washing out the drug (**Figure 8C** [↗](#) and **D** [↗](#)). In fact, the resting membrane potential became more positive (-58.2 ± 2 mV) after removal of retigabine. Together, these findings support the hypothesis that a reduction of KCNQ channel activity is responsible for the hyperexcitability of sympathetic motor neurons in aged animals.

Discussion

This research investigates the cellular and molecular mechanisms underlying age-associated sympathetic overactivity. Our results support the idea that, alongside age-related central changes, the peripheral component of the sympathetic autonomic reflex is also affected by aging. Key findings are that aging influences the intrinsic membrane properties of sympathetic motor neurons in the following ways: 1) older motor neurons exhibit a more positive RMP and increased rheobase, 2) the percentage of motor neurons displaying spontaneous activity increases with age, 3) older motor neurons respond with higher firing rates to electrical stimulation, 4) older motor neurons show a predominant tonic firing subpopulation, 5) older neurons exhibit reduced KCNQ potassium current, 6) spontaneous activity in aged neurons can be reversed using pharmacological activation of KCNQ channels.

Age-related changes in neuronal excitability

The decline in nervous system function during healthy aging has been attributed to alterations in the intrinsic membrane properties of neurons and glial cells (**Table 1** [↗](#)). These changes in electrical behavior have been observed in various experimental models, ranging from simple organisms like *C. elegans* and *Drosophila* to more complex ones like rodents and monkeys. Notably, age-related hyperexcitability has emerged as a predominant characteristic in neurons across different brain regions (**Table 1** [↗](#)).

Research on this topic has emphasized the central nervous system. However, recent evidence suggests that aging also impacts the intrinsic properties of peripheral sensory neurons leading to hyperexcitability [Hanani et al. \(2023\)](#) [↗](#). In line with this observation, our research reveals increased excitability in old sympathetic motor neurons. Collectively, these findings underscore that aging affects the intrinsic properties of peripheral neurons, challenging the notion that age-related changes are limited to the central nervous system. Hence, it is crucial to reconsider our understanding of aging-related sympathetic overactivity with a holistic perspective. Investigating how aging affects the intrinsic properties of sensory, central, and sympathetic motor neurons throughout a lifetime will be essential research to comprehend the underlying cause-and-effect mechanisms.

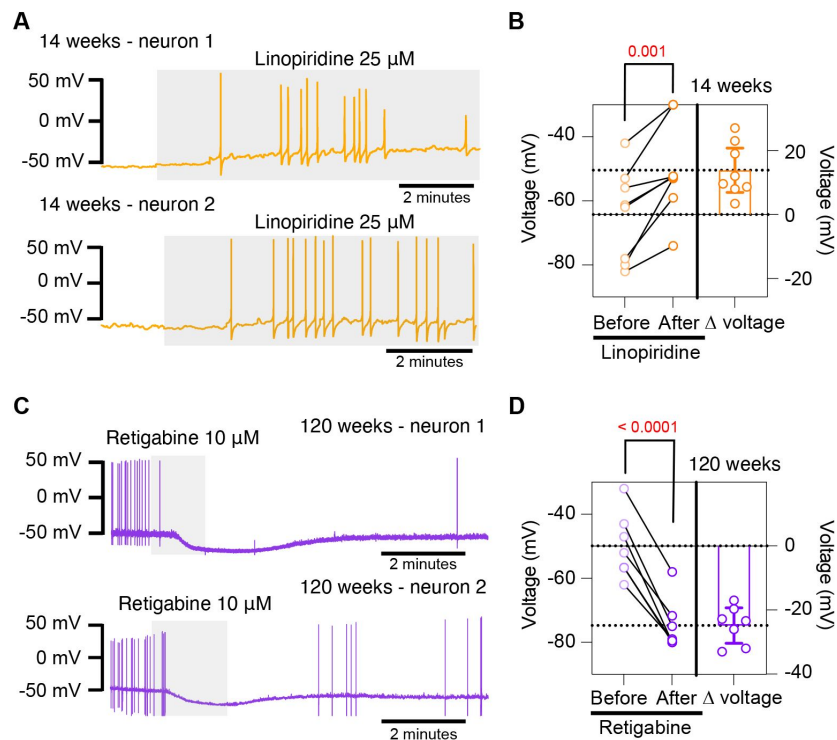


Figure 8.

Pharmacological inhibition and activation of KCNQ channels mimic the age-dependent phenotype.

(A) Membrane potential recordings from two young neurons treated with 25 μ M linopiridine during the time illustrated by the light gray box. No holding current was applied. (B) Left: Summary of the resting membrane potential measured before (light orange) and after (dark orange) the application of linopiridine. Right: summary of the depolarization produced by linopiridine calculated by subtracting the post-drug voltage from the pre-drug voltage (Δ V). Data points are from N = 2 animals, n = 8 cells, 14-week-old mice. (C) Membrane potential recordings from two aged neurons treated with 10 μ M retigabine during the time illustrated by the light gray box. No holding current was applied. (D) Left: Summary of the resting membrane potential measured before (light purple) and after (dark purple) the application of retigabine. Right: summary of the hyperpolarization produced by retigabine calculated by subtracting the post-drug voltage from the pre-drug voltage (Δ V). Data points are from N = 2 animals, n = 7 cells, 120-week-old mice. p-values are shown at the top of the graphs.

Excitability	Intrinsic properties	Ion channel/mechanisms	Cell type	Research model	Reference
Decrease	RMP-unaffecte Input resistance-unaffecte Rheobase-increase	Decrease of cholinergic response	CA1 pyramidal neurons Slices	Sprague-Daley rats 3-4 vs 25-32 months old	Potier B et al, Neuroscience, 1992
Decrease	RMP-unaffecte Input resistance-unaffecte Rheobase-increase	-	Neostriatal neurons Slices	Fischer 344 rats 3-5 vs 24-26 months old	Cepeda C et al, Neuroscience, 1992
-	RMP-unaffecte Input resistance-unaffecte Rheobase-unaffecte	Decrease of cholinergic response	CA1, CA3 and fascia dentata	F-344 rats 3 weeks vs 9 months vs 24-27 months old	Shen J and Barnes CA, Neurobiology of Aging, 1996
-	-	Increase of L-type Ca_v channel activity	CA1 hippocampal neurons Zipper slices	F-344 rats 3-6 vs 12-14 vs 23-26 months old	Thibault O and Landfield PW, Science, 1996
-	-	Decrease of NMDA protein levels	CA1 and CA3 hippocampus	Sprague-Dawley rats 3-9 vs 12-17 vs 18-24 vs 25-28 vs 29-31 months old	Wenk GL and Barnes CA, Brain research, 2000
Unaffected	RMP-unaffecte Input resistance-increase	-	Dentate granule cells Slices	Rhesus monkeys 11 vs 24 years old	Luebke JI and Rosene DL, JCN, 2003
Increase	RMP-unaffecte Input resistance-increase Rheobase-decrease	-	Layer 2/3 pyramidal neurons/ prefrontal cortex	Rhesus monkeys 8 vs 22 years old	Chang YM et al, Cerebral Cortex, 2005
Decrease	-	Increase of cAMP signaling and KCNQ/HCN activity	Dorsolateral prefrontal cortical/DELAY neuron slices	Rhesus monkeys 7-9 vs 12-13 vs 17-21 years old	Wang M et al, nature, 2011
-	-	Decrease of KCNQ expression	RT-PCR-whole brain	Drosophila 5, 25, 40, 50 and 60 days of age	Cavaliere S et al, Plos One, 2013
Increase	RMP-unaffecte Input resistance-increase	-	L3 pyramidal neurons/ lateral prefrontal cortex Slices	Rhesus monkeys 8 vs 22 years old	Coskren PJ et al, JCN, 2015
-	RMP-more negative Input resistance-decreases	Increase of inward and outward rectifier K^+ currents	Microglia in striatum, neocortex and entorhinal cortex Slices	C57BL6 mice 2-3 vs 19-24 months old	Schilling T and Eder C, Glia, 2015
Increase	RMP-unaffecte Input resistance-unaffecte	Increase of $K_v4.2/K_v4.3$ expression	CA3 hippocampal neurons Slices	F1 hybrid Fischer 344 x Brown Norway rats 2-5 vs 29-32 months old	Simkin D et al, J Neurosci, 2015
Increase	-	Decrease of BK current	Dorsal root ganglion neuron	Wistar male rats 3 and 18 months	Yu W et al, J Neurol Sci, 2015
Increase	-	Increase of glutamate-gated chloride and L-type calcium channels function		Caenorhabditis elegans	Zullo JM et al, Nature, 2019
Increase	RMP-unaffecte Rheobase-decreases	Increase of I_h	L5 pyramidal neurons Somatosensory cortex Slices	Mice C57BL/6 2-6 vs 18-29 months old	Popescu IR et al, Neurobiology of aging, 2021
Increase	RMP-unaffecte Input resistance-increase Rheobase-decrease	-	CA3 hippocampal interneurons	Sprague Dawley 5 vs 24 months old	Griego E and Galván EJ, Neurobiology of Aging, 2022
Increase	RMP-more positive Input resistance-unaffecte Rheobase-unaffecte	Decrease of KCNQ2/3 current	hypocretin/orexin neurons Hypothalamus	C57BL/6J 2-3 vs 18 months old	Li SB et al, Science, 2022
Increase	RMP-more positive Input resistance-increases Rheobase-decreases	Increase of glial-neuronal coupling	Dorsal root ganglion	Balb/c mice 3 vs 12 vs 17 months old	Hanani et al, Int J Mol Sci, 2023

Table 1.

Effects of aging on neuronal excitability.

The table compares the changes in intrinsic properties and the underlying mechanisms. It also lists cell types, research models, and references.

The molecular mechanism underlying the age-related excitability changes

The expression, distribution, regulation, and function of ion channels play a pivotal role in determining the intrinsic membrane properties of neurons. The mechanism underlying changes in age-associated excitability encompass alterations in cholinergic and glutamatergic responses and the function and expression of voltage-gated ion channels, such as Ca_V , K_V , and HCN. In recent years, research on aging has revealed a significant impact of ion channels in the aging process. For example, repressing glutamate-gated chloride channels and L-type channels extends longevity in *C. elegans* [Zullo et al. \(2019\)](#). Furthermore, inhibiting A-type K^+ channels has shown the potential to revert the intrinsic excitability of aged CA3 pyramidal neurons to a young-like state in rats [Simkin et al. \(2015\)](#).

Our data show a decrease in KCNQ current and suggest that it is a key mechanism behind the development of a more depolarized resting membrane potential and hyperexcitability in old sympathetic motor neurons. Work done in other systems has also underscored the importance of KCNQ current in the development of age-associated neuronal dysfunction. In *Drosophila*, aging is associated with reduced KCNQ expression in the brain, while KCNQ overexpression in mushroom body neurons reverses age-related memory impairment [Cavaliere et al. \(2013\)](#). Similarly, in macaques, blocking KCNQ channels partially restores memory-related firing of aged neurons to more youthful levels [Wang et al. \(2011\)](#). Recent findings by Li et al. demonstrate that aged hypocretin/orexin neurons exhibit hyperexcitability with lower KCNQ expression. Their study shows that selectively disrupting *Kcnq2/3* genes in young hypocretin neurons is sufficient to depolarize these neurons and cause sleep fragmentation, mimicking the sleep instability observed in aged mice [Li et al. \(2022\)](#). In our work, we also analyzed the effect of aging on other currents without seeing significant differences, suggesting that KCNQ channels are particularly susceptible to a process occurring during aging.

While a decrease in the expression of KCNQ during aging has been suggested as a main mechanism, we observed an increase in the abundance of KCNQ2 protein. Nevertheless, the function was diminished in old neurons, suggesting that a compensatory mechanism is being activated in sympathetic motor neurons. It is plausible that aging affects the trafficking of KCNQ channels or the insertion at the plasma membrane. Another alternative is that aging alters cofactors of the channel, such as $\text{PtdIns}(4,5)\text{P}_2$ or even the stability at the plasma membrane required for its proper opening. This mechanism remains elusive and open for exploration.

Limitations and Conclusion

The hypothesis of age-associated hyperexcitability of sympathetic motor neurons should be tested in the neurons of other sympathetic ganglia, including the stellate and celiac. While our study allowed us to determine the effect of aging on sympathetic neurons in isolation from other components of the sympathetic reflex, it did not provide insight into whether the hyperexcitability of motor neurons is a compensatory mechanism responding to earlier changes in the brain or target tissues. This question has posed a long-standing challenge, especially in various pathologies where sympathetic overactivity, such as arrhythmias and hypertension, play a role. In such cases, sympathetic overactivity could result from deterioration in the target organs or sensory components, such as the heart and carotid body. Further research and exploration are needed to unravel these complex interactions and establish a comprehensive understanding of the mechanisms underlying age-related changes in the sympathetic nervous system.

In conclusion, this study demonstrates that aging directly impacts the intrinsic electrical properties of sympathetic motor neurons. Furthermore, our research postulates that the decrease in KCNQ function underlies the hyperexcitability of sympathetic motor neurons. These findings

shed light on the mechanisms involved in age-related changes within the sympathetic nervous system and offer a promising avenue for further investigation and potential intervention.

Methods and Materials

Animal Models

Male C57BL/6 WT mice were purchased from the Jackson Laboratory (12 weeks, RRID:IMSR_JAX:000664) or obtained from the NIA-NIH colony (ages 64 weeks and 115 weeks). All animals were kept in an animal facility with controlled conditions and were given standard chow and water *ad libitum*. The animal handling protocol was approved by the University of Washington Institutional Animal Care and Use Committee.

Sympathetic motor neuron cell culture

Neurons from superior cervical ganglia (SCG) were prepared by enzymatic digestion following a standardized protocol for rats [Vivas et al. \(2014\)](#) but reducing the enzyme concentration in half. Isolated neurons were plated on poly-L-lysine (#Cat: P1524-25MG, Sigma) coated glass coverslips and incubated in 5% CO₂ at 37°C in DMEM supplemented with 10% FBS and 0.2% penicillin/streptomycin.

Imaging of neurite growth

Cells were imaged in culture medium at 24 and 72 h and one week after isolation for neurites analysis. Images were taken using the bright field of an LSM 880 confocal microscope (Zeiss).

Electrophysiological Recordings

Voltage responses were recorded using the perforated-patch configuration in the current-clamp mode, whereas KCNQ currents were recorded using the whole-cell configuration in voltage-clamp mode. We used an Axopatch 200B amplifier coupled with an Axon Digidata 1550B data acquisition board (Molecular Devices Electrophysiology) and HEKA EPC 9 amplifier (HEKA Elektronik) to acquire the electrical signals. Patch pipettes had a resistance of 2 – 4 MΩ. A liquid junction potential of 4 mV was calculated using the pCLAMP 10 software and not corrected while recording. Instead, V_{rest} reported in the Results section was corrected during analysis. Voltage responses were sampled at 5 KHz, whereas currents were sampled at 2 KHz. For current recordings in voltage clamp mode, cell capacitance was canceled, and series resistances of <10 MΩ were compensated by 70%. The voltage error due to any remaining series resistance is expected to be <4 mV. The bath solution (Ringer's solution) contained 150 mM NaCl, 2.5 mM KCl, 2 mM CaCl₂, 1 mM MgCl₂, 10 mM HEPES, and 8 mM glucose, adjusted to pH 7.4 with NaOH. The internal solution used to fill the whole-cell pipettes contained 175 mM KCl, 1 mM MgCl₂, 5 mM HEPES, 0.1 mM K₄BAPTA, 3 mM Na₂ATP, and 0.1 mM Na₃GTP, adjusted to pH 7.2 with KOH. For current-clamp recordings by perforated patch, 60 amphotericin B (Cat #: A4888) was added to the pipette solution to facilitate electrical access to the cell. The bath solution was perfused at 2 ml/minute, permitting solution exchange surrounding the recording cell with a time constant of 4 s. Sodium currents were recorded 2-8 h after isolation using a low-sodium ringer.

Protein extraction and abundance determination

Protein from sympathetic ganglia was harvested in RIPA buffer (#89900, Thermo Scientific) with Complete, Mini, EDTA-free protease inhibitor cocktail (#11836170001, Roche) for 15 min at 4°C. Post-nuclear supernatant was isolated by centrifuging for 20 minutes at 13,600 g at 4°C. Protein concentration was quantified on a plate reader using the Pierce BCA protein assay kit (#23225, Thermo Scientific). Gel lanes were loaded with 30 µg of total protein. Protein samples were resolved in 4-12% Bis-Tris gels under reducing conditions. Proteins were transferred onto

nitrocellulose membranes (0.2 μ m; #LC2000, Life Technologies) using the Mini-Bolt system (#A25977, Thermo Scientific). Membranes were blotted using rabbit anti-KCNQ2 (ab22897, Abcam, RRID: AB_775890, 1:500). Blotted bands were detected using HRP conjugated secondary antibodies goat anti-rabbit conjugated with HRP (#1706515, Bio-Rad, 1:15, 000). ImageJ was used to calculate the fluorescence density of each band. The abundance of KCNQ2 was reported as normalized to total protein and relative to the abundance in tissue from young animals.

Reagents

XE-991 (Cat #: X-101), Phrixotoxin-1 (Cat #: STP-700), Guangtoxin-1E (Cat #: STG-200) and Tetrodotoxin (T-550) were obtained from Alomone Labs.

Data analysis and statistics

We used IGOR Pro (IGOR Software, WaveMetrics, RRID: SCR_000325), Excel (Microsoft), and Prism (GraphPad, RRID: SCR_002798) to analyze data. ImageJ (RRID: SCR_003070) was used to process images. Data were collected from independent experiments from at least three mice and are presented as Mean $\pm$ SEM. The statistical analyses were performed using the parametric Student's t-test when comparing two variables and an ordinary one-way ANOVA (Dunnett's multiple comparisons test) when comparing three or more variables. Graph Pad was used to calculate Pearson correlation coefficients using one-tail analysis. A non-parametric statistical test (Mann-Whitney Wilcoxon) was used to test for statistical significance between the percentage of firing and non-firing neurons. p values <0.05 as statistical significance. The number of cells used for each experiment is detailed in each figure legend.

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Reviewer #1 (Public Review):

Summary:

The authors study age-related changes in the excitability and firing properties of sympathetic neurons, which they ascribe to age-related changes in the expression of KCNQ (Kv7, "M-type") K⁺ currents in rodent sympathetic neurons, whose regulation by GPCRs has been most thoroughly studied for over 40 years.

Strengths:

The strengths include the rigor of the current-clamp and voltage-clamp experiments and the lovely, crisp presentation of the data. The separation of neurons into tonic, phasic and adapting classes is also interesting, and informative. The ability to successfully isolate and dissociate peripheral ganglia from such older animals is also quite rare and commendable! There is much useful detail here.

Weaknesses:

Whereas the description of the data are very nice and useful, the manuscript does not provide much in the way of mechanistic insights. As such, the effect is more of an epiphenomenon of unclear insight, and the authors cannot ascribe changes in signaling mechanisms, such as that of M1 mAChRs to the phenomena that is supported by data.

<https://doi.org/10.7554/eLife.91663.2.sa2>

Reviewer #2 (Public Review):

Summary:

This research shows compelling and detailed evidence showing that aging influences intrinsic membrane properties of peripheral sympathetic motor neurons, which become hyperexcitable. The authors found that sympathetic motor neurons from old mice exhibit increased firing rates (spontaneous and evoked), more depolarized membrane resting potential, and increased rheobase. Furthermore, the study investigates cellular mechanisms underlying age-associated hyperexcitability and shows solid evidence supporting that a decreased activity of KCNQ channels during aging is a major contributor to the increased excitability of sympathetic old neurons. All conclusions of this paper are well supported by the data.

Strengths:

Detailed and rigorous analysis of electrical responses of peripheral sympathetic motor neurons using electrophysiology (perforated patch and whole-cell recordings). The study

identifies a decrease in KCNQ current as a cellular mechanism behind age-induced hyperexcitability in sympathetic motor neurons.

Weaknesses:

None, the revised version of the manuscript has addressed all my concerns.

<https://doi.org/10.7554/eLife.91663.2.sa1>

Reviewer #3 (Public Review):

This study described changes in membrane excitability and Na⁺ and K⁺ current amplitudes of sympathetic motor neurons in culture. The findings indicate that neurons isolated from aged animals show increased membrane excitability manifested as increased firing rates in response to electrical stimulation and changes in related membrane properties including depolarized resting membrane potential, increased rheobase, and spontaneous firing. By contrast, neuron cultures from young mice show little to no spontaneous firing and relatively low firing rates in response to current injection. These changes in excitability correlate with reductions in the magnitude of KCNQ currents in neurons cultured from aged mice compared to neurons from cultured from young mice. The authors conclude that aging promotes hyperexcitability of sympathetic motor neurons through changes in KCNQ channels.

The electrophysiological cataloging of the neuronal properties is well done, and the experiments are performed using perforated patch recordings which preserves the internal constituents of neurons, providing confidence that the effects seen are not due to washout of regulators from the cells. The main weakness is that this study is a descriptive tabulation of changes in the electrophysiology of neurons in culture, and the effects shown are correlative rather than establishing causality. Pharmacological support is provided indicating that blockade or enhancement of KCNQ reverses the changes in excitability, but the specifics of the effects and relevance to intact preparations are unclear. Additional experiments in slice cultures would provide greater significance on the potential relevance of the findings for intact preparations.

<https://doi.org/10.7554/eLife.91663.2.sa0>

Author response:

The following is the authors' response to the current reviews.

Reviewer #1:

Summary:

The authors study age-related changes in the excitability and firing properties of sympathetic neurons, which they ascribe to age-related changes in the expression of KCNQ (Kv7, "M-type") K⁺ currents in rodent sympathetic neurons, whose regulation by GPCRs has been most thoroughly studied for over 40 years.

Strengths:

The strengths include the rigor of the current-clamp and voltage-clamp experiments and the lovely, crisp presentation of the data, The separation of neurons into tonic, phasic and adapting classes is also interesting, and informative. The ability to successfully isolate and dissociate peripheral ganglia from such older animals is also quite rare and commendable! There is much useful detail here.

Thank you for recognizing the effort we put on presenting the data and analyzing the neuronal populations. I also believe the ability to isolate neurons from old animals is worth communicating to the scientific community.

Weaknesses:

Where the manuscript becomes less compelling is in the rapamycin section, which does not provide much in the way of mechanistic insights. As such, the effect is more of an epiphenomenon of unclear insight, and the authors cannot ascribe a signaling mechanism to it that is supported by data. Thus, this latter part rather undermines the overall impact and central advance of the manuscript. The problem is exacerbated by the controversial and anecdotal nature of the entire mTor/aging field, some of whose findings have very unfortunately had to be recently retracted.

I would strongly recommend to the authors that they end the manuscript with their analysis of the role of M current/KCNQ channels in the numerous age-related changes in sympathetic neuron function that they elegantly report, and save the rapamycin, and possible mTor action, for a separate line of inquiry that the authors could develop in a more thorough and scholarly way.

Whereas the description of the data are very nice and useful, the manuscript does not provide much in the way of mechanistic insights. As such, the effect is more of an epiphenomenon of unclear insight, and the authors cannot ascribe changes in signaling mechanisms, such as that of M1 mAChRs to the phenomena that is supported by data.

I appreciate the new comment. We had agreed that our rapamycin experiments did not allow to ascribe the mechanism to the signaling pathway of mTOR. The new comment mentions M1 mAChRs signaling as another potential signaling mechanism. Our work centered on determining whether aging altered the function of sympathetic motor neurons and defining the mechanism. We presented evidence showing that the mechanism is a reduction of the M-current. We did not attempt to identify the signaling mechanism linking aging to a reduction in M-current. Therefore, we agree with the reviewer that we do not provide further details on the mechanism and that that remains an open question. However, I find it harsh to say that “the effect is more of an epiphenomenon of unclear insight”. How could we possibly test that the effect of aging on the excitability of these neurons only arises as a secondary effect or that is not causal? How could we test for sufficiency and necessity of aging? How could we modify the state of aging to test for causality? We would have to reverse aging and show that the effect on the excitability is gone. And that is exactly what we tried to do with the rapamycin experiment.

Reviewer #1 (Recommendations For The Authors):

(1) The significance values greater than $p < 0.05$ do not add anything and distract focus from the results that are meaningful. Fig. 5 is a good example. What does $p = 0.7$ mean? Or $p = 0.6$? Does this help the reader with useful information?

I thank Reviewer 1 for raising this question. We have attempted different versions of how we report p values, as we want to make sure to address rigor and transparency in reporting data. As corresponding author, I favor reporting p values for all statistical comparisons. To help the reader identifying what we considered statistically significant, we color coded the p values, with red for $p\text{-value} < 0.05$ and black for $p\text{-value} > 0.05$. As a reader, seeing a $p\text{-value} = 0.7$ allows me to know that the authors performed an analysis comparing these conditions and found the mean not to be different. Not presenting the p-value makes me wonder whether the authors even analyzed those groups. In other words, I value more the ability to analyze the

data seeing all p-values than not being distracted by not-significant p-values. This is just my preference.

(2) *Fig. 1 is not informative and should be removed.*

I thank Reviewer 1 for the suggestion. In previous drafts of the manuscript, this figure was included only as a panel. However, we decided it was better to guide the reader into the scope of our work. This is part of our scientific style and, therefore, we prefer to keep the figure.

(3) *The emphasis on a particular muscarinic agonist favored by many ion channel physiologists, oxotremorine, is not meaningful (lines 192, 198). The important point is stimulation of muscarinic AChRs, which physiologically are stimulated by acetylcholine. The particular muscarinic agonist used is unimportant. Unless mandated by eLife, "cholinergic type 1 muscarinic receptors" are usually referred to as M1 mAChRs, or even better is "Gq-coupled M1 mAChRs." I don't think that Kruse and Whitten, 2021 were the first to demonstrate the increase in excitability of sympathetic neurons from stimulation of M1 mAChRs. Please try and cite in a more scholarly fashion.*

A) I have modified lines 192 and 198 removing mention to oxotremorine.

B) I have modified the nomenclature used to refer to cholinergic type 1 muscarinic receptors.

C) I cited references on the role of M current on sympathetic motor neuron excitability. I also removed the reference (Kruse and Whitten, 2021) referring only on the temporal correlation between the decrease of KCNQ current with excitability.

(4) *The authors may want to use the term "M current" (after defining it) as the current produced by KCNQ2&3-containing channels in sympathetic neurons, and reserve "KCNQ" or "Kv7" currents as those made by cloned KCNQ/Kv7 channels in heterologous systems. A reason for this is to exclude currents KCNQ1-containing channels, which most definitely do not contribute to the "KCNQ" current in these cells. I am not mandating this, but rather suggesting it to conform with the literature.*

Thank you for the suggestion. I have modified the text to use the term M current. I maintain the use of KCNQ only when referring to KCNQ channel, such as in the section describing the abundance of KCNQ2.

(5) *The section in the text on "Aging reduces KCNQ current" is confusing. Can the authors describe their results and their interpretation more directly?*

I am not sure to understand the request. I assumed point 5 and 6 are related and decided to answer point 6.

(6) *Please explain the meaning of the increase in KCNQ2 abundance with age in Fig. 6G. How is this increase in KCNQ2 expression consistent with an increase in excitability? The explanation of "The decrease in KCNQ current and the increase in the abundance of KCNQ2 protein suggest a potential compensatory mechanism that occurs during aging, which we are actively investigating in an independent study." is rather odd, considering that the entire thesis of this paper is that changes in excitability and firing properties are underlied by changes in KCNQ2/3 channel expression/density. Suddenly, is this not the case?? What about KCNQ3? It would be very enlightening if the authors would just quantify the ratio of KCNQ2:KCNQ3 subunits in M-type channels in young and old mice using simple TEA dose/response curves (see Shapiro et al., JNS, 2000; Selyanko et al., J. Physiol., Hadley et al., Br. J. Pharm., 2001 and a great many more). It is also surprising*

that the authors did not assess or probe for differences in mAChR-induced suppression of M current between SCG neurons of young and old mice. This would seem to be a fundamental experiment in this line of inquiry.

A. Please explain the meaning of the increase in KCNQ2 abundance with age in Fig. 6G. How is this increase in KCNQ2 expression consistent with an increase in excitability? The explanation of "The decrease in KCNQ current and the increase in the abundance of KCNQ2 protein suggest a potential compensatory mechanism that occurs during aging, which we are actively investigating in an independent study." is rather odd, considering that the entire thesis of this paper is that changes in excitability and firing properties are underlied by changes in KCNQ2/3 channel expression/density. Suddenly, is this not the case?? Our interpretation is that the decrease in M current is not caused by a decrease in the abundance of KCNQ (2) channels. We do not claim that changes in excitability are underlied by a reduction in the expression or density of KCNQ2 channels. On the contrary, our working hypothesis is that the reduction in M current is caused by changes in traffic, degradation, posttranslational modifications, or cofactors for KCNQ2 or KCNQ3 channels. We have modified the description in the results section to clarify this concept.

B. What about KCNQ3? Unfortunately, we did not find an antibody to detect KCNQ3 channels. I have added a sentence to state this.

C. KCNQ2:KCNQ3 subunits in M-type channels in young and old mice using simple TEA dose/response curves. This is a great idea. Thank you for the suggestion. Is this a necessary experiment for the acceptance of this manuscript?

D. It is also surprising that the authors did not assess or probe for differences in mAChR-induced suppression of M current between SCG neurons of young and old mice. This would seem to be a fundamental experiment in this line of inquiry. Reviewer 1 is correct. We did not assess for differences in the suppression of M current by mAChR activation. We do not see the connection of this experiment with the scope of the current investigation.

(7) Why do the authors use linopirdine instead of XE-991? Both are dirty drugs hardly specific to KCNQ channels at 25 uM concentrations, but linopirdine less so. The Methods section lists the source of XE991 used in the study, not linopirdine. Is there an error?

A. Why do the authors use linopirdine instead of XE-991? After validation of KCNQ2/3 inhibition by Linopirdine, we found the effect on membrane potential recordings to be reproducible. Linopirdine has also been reported to be reversible. We wanted to assess reversibility on the excitability of young neurons. We did not find the effect to be reversible. We performed experiments applying XE-991 while recording the membrane potential. XE-991 did not show a clear effect. I was not surprised by this. It is very likely that the pharmacological inhibition of one channel leads to the activation of other channel types. This is highlighted in the work by Kimm, Khaliq, and Bean, 2015. "Further experiments revealed that inhibiting either BK or Kv2 alone leads to recruitment of additional current through the other channel type during the action potential as a consequence of changes in spike shape." In fact, it was quite remarkable that the aged and young phenotypes were mimicked by targeting KCNQ pharmacologically.

B. Both are dirty drugs hardly specific to KCNQ channels at 25 uM concentrations, but linopirdine less so. I have added a sentence to point out that linopirdine is less potent than XE-991. It reads: "We want to point out that linopirdine is less potent than XE-991 and that it has been reported to activate TRPV1 channels (Neacsu and Babes, 2010). Despite this limitation, the application of linopirdine to young sympathetic motor neurons led to depolarization and firing of action potentials."

C. The Methods section lists the source of XE991 used in the study, not linopirdine. Is there an error? Thank you for pointing out this. I have added information for both retigabine and linopirdine in the Methods section, both were missing.

(8) Can the authors use a more scientific explanation of RTG action than "activating KCNQ channels?" For instance, RTG induces both a negative-shift in the voltage-dependence of activation and a voltage-independent increase in the open probability, both of which differing in detail between KCNQ2 and KCNQ3 subunits. The authors are free to use these exact words. Thus, the degree of "activation" is very dependent upon voltage at any voltages negative to the saturating voltages for channel activation.

I have modified the text to reflect your suggestion.

(9) Methods: did the authors really use "poly-L-lysine-coated coverslips?" Almost all investigators use poly-D-lysine as a coating for mammalian tissue-culture cells and more substantial coatings such as poly-D-lysine + laminin or rat-tail collagen for peripheral neurons, to allow firm attachment to the coverslip.

That is correct. We used poly-L-lysine-coated coverslips. Sympathetic motor neurons do not adhere to poly-D-Lysine.

(10) As a suggestion, sampling M-type/KCNQ/Kv7 current at 2 kHz is not advised, as this is far faster than the gating kinetics of the channels. Were the signals filtered?

It is correct. Currents were sampled at 2KHz. Data were low-pass filtered at 3 KHz. Our conditions are not far from what is reported by others. Some sample at 10KHz and even 50 KHz. Others do not report the sample frequency.

Reviewer #2:

Weaknesses:

None, the revised version of the manuscript has addressed all my concerns.

I am glad we were able to satisfy previous concerns.

Reviewer #3:

The main weakness is that this study is a descriptive tabulation of changes in the electrophysiology of neurons in culture, and the effects shown are correlative rather than establishing causality.

Allow me to clarify our previous responses and determine how this aligns with your concerns. In the previous revision, Reviewer 3 wrote: "It is difficult to know from the data presented whether the changes in KCNQ channels are in fact directly responsible for the observed changes in membrane excitability." And suggested to "use of blockers and activators to provide greater relevance." I assumed these comments were the main concern and that doing such experiments was enough to satisfy the criticism. It is discouraging to see that our experiments did not satisfy the concerns of the reviewer of being correlative.

If Reviewer 3 is referring to establishing causality between aging and a reduction in M current, I would like to emphasize that such endeavor is complicated as there is not a clear experiment to solve that issue. Our best attempt was to reverse aging with rapamycin, but the recommendation was to remove those experiments.

... but the specifics of the effects and relevance to intact preparations are unclear. Additional experiments in slice cultures would provide greater significance on the potential relevance of the findings for intact preparations.

I apologize for missing this point in the previous revision. The proposed experiments will require an upward microscope coupled to an electrophysiology rig. Unfortunately, I do not have the equipment to do these experiments.

Summary of recommendations from the three reviewers:

Please make corrections as suggested by reviewer 1 to improve the manuscript. Specifically, reviewer 1 suggests making changes to p values in Figure 5,

It is not clear what the suggested changes are. The comment from Reviewer 1 says: The significance values greater than $p < 0.05$ do not add anything and distract focus from the results that are meaningful. If the suggested change is to remove p values > 0.05 , I have explained my rational for keeping those values. If the Journal has a specific format on how to report p -values, I will be happy to make appropriate changes.

and the importance of citing original scholarly works related to effects of increase in excitability of sympathetic neurons by M1 receptors, and the terminology for M currents and KCNQ currents. These changes will improve the manuscript and are strongly recommended.

I cited original papers on that area, and changed the terminology for M current. I kept KCNQ when referring to the channel protein or abundance.

The section dealing with Aging Reduces KCNQ currents seems to contain a lot of extraneous information especially in the last part of the long paragraph and this section should be rewritten for improved clarity... and - the implications or lack thereof - of the correlation of KCNQ with AP firing rates.

A. I removed extraneous information in that section. It now reads: Previous work by our group and others demonstrated that cholinergic stimulation leads to a decrease in M current and increases the excitability of sympathetic motor neurons at young ages \cite{RN67,RN68,RN69,RN71, RN72, RN73, RN74, RN75}. The molecular determinants of the M current are channels formed by KCNQ2 and KCNQ3 in these neurons \cite{RN76, RN77, RN70}. Thus, Figure 6A shows a voltage response (measured in current-clamp mode) and a consecutive M current recording (measured in voltage-clamp mode) in the same neuron upon stimulation of cholinergic type 1 muscarinic receptors. It illustrates the temporal correlation between the decrease of M current with the increase in excitability and firing of APs upon activation with oxotremorine. This strong dependence led us to hypothesize that aging decreases M current, leading to a depolarized RMP and hyperexcitability (Figure 6B). For these experiments, we measured the RMP and evoked activity using perforated patch, followed by the amplitude of M current using a whole-cell voltage clamp in the same cell. We also measured the membrane capacitance as a proxy for cell size. Interestingly, M current density was smaller by 29% in middle age (7.5 ± 0.7 pA/pF) and by 55% in old (4.8 ± 0.7 pA/pF) compared to young (10.6 ± 1.5 pA/pF) neurons (Figure 6C-D). The average capacitance was similar in young (30.8 ± 2.2 pF), middle-aged (27.4 ± 1.2 pF), and old (28.8 ± 2.3 pF) neurons (Figure 6E), suggesting that aging is not associated with changes in cell size of sympathetic motor neurons, and supporting the hypothesis that aging alters the levels of M current. Next, we tested the effect on the abundance of the channels mediating M current. Contrary to our expectation, we observed that KCNQ2 protein levels were 1.5 ± 0.1 -fold higher in old

compared to young neurons (Figure 6F-G). Unfortunately, we did not find an antibody to detect consistently KCNQ3 channels. We concluded that the decrease in M current is not caused by a decrease in the abundance of KCNQ2 protein.

B. and - the implications or lack thereof - of the correlation of KCNQ with AP firing rates. I am not sure to understand the request on the section of the correlation of KCNQ with AP firing rate. I divided the long paragraph.

The apparent lack of correlation between KCNQ current and KCNQ2 protein needs to be better explained. This is a central part of the study and this result undercuts the premise of the paper.

Indeed, total KCNQ2 protein abundance increases while M current decreases. We do not claim in our work that changes in excitability are caused by a reduction in the expression or density of KCNQ2 channels. On the contrary, our current working hypothesis is that the reduction in M current is caused by changes in traffic, degradation, posttranslational modifications, or cofactors for KCNQ2 or KCNQ3 channels. I have modified the description in the results section and discussion to clarify this concept.

Additionally, the poor specificity of Linordipine for KCNQ should be pointed out in the limitations.

I pointed this limitation. It reads: We want to point out that linopirdine is less potent than XE-991 and that it has been reported to activate TRPV1 channels (Neacsu and Babes, 2010). Despite this limitation, the application of linopirdine to young sympathetic motor neurons led to depolarization and firing of action potentials.

Finally, the editor notes that the author response should not contain ambiguities in what was addressed in the revision. In the original summary of consolidated revisions that were requested, one clearly and separately stated point (point 4) was that experiments in slice cultures should be strongly considered to extend the significance of the work to an intact brain preparation. The author response letter seems to imply that this was done, but this is not the case. The author response seems to have combined this point with another separate point (point 3) about using KCNQ drugs, and imply that all concerns were addressed. Authors should be clear about what revisions were in fact addressed.

As corresponding author, and direct responsible of the document provided for the reply to the reviewers, I apologize for my mistake. After reviewing this comment, I realized I did not respond to the Major points in the section of the Recommendations for the authors from Reviewer 3. I missed that entire section. My previous responses addressed the Public review of reviewer 3. When doing so, I did not separate the sentences, omitting the request on performing the experiment in slices.

The following is the authors' response to the original reviews.

Reviewer #1

Summary:

The authors study age-related changes in the excitability and firing properties of sympathetic neurons, which they ascribe to age-related changes in the expression of KCNQ (Kv7, "M-type") K⁺ currents in rodent sympathetic neurons, whose regulation by GPCRs has been most thoroughly studied for over 40 years. The authors suggest the ingestion of rapamycin may partially reverse the age-related decrease in M-channel

expression. With the rapamycin part included, it is unclear how this work will impact the field of age-related neuronal dysfunction, as the mechanistic information is not strong.

Strengths:

The strengths include the rigor of the current-clamp and voltage-clamp experiments, the lovely, crisp presentation of the data, and the expert statistics. The separation of neurons into tonic, phasic, and adapting classes is also interesting, and informative. The writing is also elegant, and crisp. The above is especially true of the manuscript up until the part dealing with the effects of rapamycin, which becomes less compelling.

We appreciate the thoughtful comments and constructive feedback to improve the impact of the manuscript.

Weaknesses:

Where the manuscript becomes less compelling is in the rapamycin section, which does not provide much in the way of mechanistic insights. As such, the effect is more of an epiphenomenon of unclear insight, and the authors cannot ascribe a signaling mechanism to it that is supported by data. Thus, this latter part rather undermines the overall impact and central advance of the manuscript. The problem is exacerbated by the controversial and anecdotal nature of the entire mTor/aging field, some of whose findings have very unfortunately had to be recently retracted.

I would strongly recommend to the authors that they end the manuscript with their analysis of the role of M current/KCNQ channels in the numerous age-related changes in sympathetic neuron function that they elegantly report, and save the rapamycin, and possible mTor action, for a separate line of inquiry that the authors could develop in a more thorough and scholarly way.

We agree with the reviewer in that we cannot ascribe a signaling mechanism to the reversibility observed with rapamycin. Therefore, we are following the recommendation of the reviewer and have removed the rapamycin section.

We want to emphasize that, in the aging field, any advancement in the knowledge of how drugs such as rapamycin reverse age-associated phenotypes is of crucial importance. These drugs, commonly referred to as aging interventions, include rapamycin, calorie restriction, elamipretide, and metformin. We could have used any of these interventions. And yet, the cellular and molecular mechanisms for each one of these anti-aging drugs are unknown.

We want to note that, although the nature of the mTOR field is controversial, the effect of rapamycin in extending lifespan and improving health is not. At least these authors have not been able to find retracted papers on that subject or notices from the NIA alerting on this issue. We kindly request the reviewer to provide the references related to rapamycin that were retracted so we can evaluate how that affects the rigor of the premise for our future work.

As authors, we also find it important to note that we are confident of our observations regarding the effect of rapamycin, and that we are not removing this section because we are retracting our claims. We will use these data to continue our research of the mechanism behind the effect of aging on sympathetic motor neurons.

Reviewer #2:

Summary:

This research shows compelling and detailed evidence showing that aging influences intrinsic membrane properties of peripheral sympathetic motor neurons such that they become more excitable. Furthermore, the authors present convincing evidence that the oral administration of the anti-aging drug Rapamycin partially reversed hyperexcitability in aged neurons. This study also investigates the molecular mechanisms underlying age-associated hyperexcitability in mouse sympathetic motor neurons. In that regard, the authors found an age-associated reduction of an outward current having properties similar to KCNQ2/Q3 potassium current. They suggested a reduction of KCNQ2/Q3 current density in aged neurons as a potential mechanism behind their overactivity.

Strengths:

Detailed and rigorous analysis of electrical responses of peripheral sympathetic motor neurons using electrophysiology (perforated patch and whole-cell recordings). Most of the conclusions of this paper are well supported by the data.

We thank the reviewer for valuing our effort to present a detailed and rigorous analysis.

Weaknesses:

(1) The identity of the age-associated reduced current as KCNQ2/Q3 is not corroborated by pharmacology (blocking the current with the specific blocker XE-991).

We have performed experiments using blockers of KCNQ channels. See responses below.

(2) The manuscript does not include a direct test of the reduction of KCNQ current as the mechanism behind age-induced hyperexcitability.

Thank you for raising this point. We have performed experiments blocking KCNQ channels with Linopiridine in young neurons and found that the pharmacological reduction of KCNQ current was enough to depolarize the cell and, in some cases, elicit the firing of action potentials. We present the results in a new figure. We also added the description in the Results section.

Reviewer #3:

This is a descriptive study of membrane excitability and Na⁺ and K⁺ current amplitudes of sympathetic motor neurons in culture. The main findings of the study are that neurons isolated from aged animals show increased membrane excitability manifested as increased firing rates in response to electrical stimulation and changes in related membrane properties including depolarized resting membrane potential, increased rheobase, and spontaneous firing. By contrast, neuron cultures from young mice show little to no spontaneous firing and relatively low firing rates in response to current injection. These changes in excitability correlate with significant reductions in the magnitude of KCNQ currents in aged neurons compared to young neurons. Treating cultures with the immunosuppressive drug, rapamycin, which has known antiaging effects in model animals appears to reverse the firing rates in aged neurons and enhance KCNQ current. The authors conclude that aging promotes hyperexcitability of sympathetic motor neurons.

The electrophysiological cataloging of the neuronal properties is generally well done, and the experiments are performed using perforated patch recordings which preserve the internal constituents of neurons, providing confidence that the effects seen are not due to washout of regulators from the cells.

The main weakness is that this study is a descriptive tabulation of changes in the electrophysiology of neurons in culture, and the effects shown are correlative rather than establishing causality. It is difficult to know from the data presented whether the changes in KCNQ channels are in fact directly responsible for the observed changes in membrane excitability.

We appreciate the constructive criticism. In an attempt to assess whether changes in KCNQ are in fact directly responsible for the changes in membrane excitability, we have performed experiments blocking KCNQ channels with Linopirdine in young neurons and found that the pharmacological reduction of KCNQ current was enough to depolarize the cell and, in some cases, elicit the firing of action potentials. Conversely, we activated KCNQ channels in old neurons with retigabine and found that the pharmacological activation was enough to hyperpolarize the membrane potential and stop the firing of action potentials. This effect was reversible. These two experiments provide solid evidence to our statement that age-associated reduction of KCNQ activity is responsible for the hyperexcited state in sympathetic motor neurons. We present the results in a new figure (Figure 8). We also added the description in the Results section.

Furthermore, a notable omission seems to be the analysis of Ca²⁺ currents which have been widely linked to alterations in membrane properties in aging.

We thank the reviewer for the comment. We did omit to include data on our studies of calcium currents. We agree that the study of the effect of calcium currents is relevant as it can influence the afterhyperpolarization. Furthermore, we believe that potential effects on calcium currents need to be studied in relation to other physiological processes that depend on calcium, including excitation-transcription coupling, calcium handling, and neurotransmitter release. Adding this information to this manuscript would only contribute to the tabulation of effects that we observe in sympathetic motor neurons with aging. As our main goal was to determine the ion channels responsible for the hyperexcited state, voltage-gated calcium channels or other calcium sources could have reflected a more indirect mechanism as compared to changes in sodium or potassium currents. We will continue our investigation on calcium currents and report our observations in the future, but for now, we have decided to leave it out of this work.

As well, additional experiments in slice cultures would provide greater significance on the potential relevance of the findings for intact preparations. Finally, experiments using KCNQ blockers and activators could provide greater relevance that the observed changes in KCNQ are indeed connected to changes in membrane excitability.

We are happy to report that we have performed these experiments and that the results strengthen the conclusion that changes in KCNQ are connected to changes in membrane excitability.

Recommendations for the authors:

We recommend the following essential revisions summarized from the reviews:

(1) Is the change in KCNQ current responsible for the altered membrane excitability? What happens to membrane excitability when KCNQ is partially blocked (see reviewer 2 comment below)? Conversely, what happens to the excitability of aged neurons if KCNQ is activated (e.g., with retigabine)? (see reviewer 3 comment below). Results of these important experiments are needed to support the argument that KCNQ underlies the alterations in firing and membrane excitability.

We have responded to this point. Thank you for the suggested experiments. In summary, the new experiments show that blocking KCNQ channels in young neurons lead to depolarization, and in some cases, the firing of action potentials. Conversely, the activation of KCNQ channels in aged neurons leads to hyperpolarization and a cease of firing. We have added a new figure and reported the results in the Results section.

(2) Rapamycin experiments are underdeveloped and weak. These should be further developed by examining the effects of KCNQ blockers to see if their effects on membrane excitability are reversed. Also, see comment 2 from reviewer 1.

We have followed the recommendation by reviewer 1 and removed the section on rapamycin.

(3) The study should examine voltage-gated calcium currents to determine potential changes in these currents with aging. See reviewer 3 comments.

We thank the reviewer for the comment. We performed preliminary experiments and found that aging impacts calcium currents. However, we omitted to include the data. In our opinion, the changes in calcium currents are outside the scope of this work, as the changes could be related to physiological processes that go beyond the control of firing. Effects on calcium currents need to be studied in relation to other physiological processes that depend on calcium, including excitation-transcription coupling, calcium handling, and neurotransmitter release. The study of the relationship between changes in calcium currents and those physiological processes would require multiple experiments and detailed analysis. We will continue our investigation on calcium currents and report our observations in the future, but for now, we have decided to leave it out of this work.

We have also edited suggestions in the Figures and Legends.

(2) In Fig.4 panel H, Y-axis must be # AP at 100 pA.

We corrected the axis in Figure 4H.

(3) In Legend Fig. 5, the number of cells for each subpopulation (n) needs to be corrected. In plots F-I, n= 9, 7, and 3 seem to be the number of adapting cells for 12-, 64- and 115w-old, respectively, instead of the number of single, phasic, and old cells for 12-week-old mice. A similar correction seems to be needed for 64-week-old and 115-week-old.

We corrected the n number in Figure 5.

(4) In Figure 6 panel C, it would be helpful for a reader to align the voltage protocol depicted with the current shown.

We have aligned the voltage protocol to the current traces.

(5) In the legend of Figure 7, the description of panel A ends with "Magnitude of voltage step to elicit each trace is shown in black", however in panel A there is no voltage depiction. In the description of panel D, "N = X animals, n=x cells" must be corrected.

We have modified the legend to clarify. It now reads: "Text at the right of each current trace corresponds to the voltage used to elicit that current."

New Figure 8

Author response image 1.

Pharmacological inhibition and activation of KCNQ channels mimic the age-dependent phenotype. A. Membrane potential recordings from two young neurons treated with 25 μM linopirdine during the time illustrated by the light gray box. No holding current was applied. B. Left: Summary of the resting membrane potential measured before (light orange) and after (dark orange) the application of linopirdine. Right: Summary of the depolarization produced by linopirdine calculated by subtracting the post-drug voltage from the pre-drug voltage (ΔV). Data points are from $N = 2$ animals, $n = 8$ cells, 14-week-old mice. C. Membrane potential recordings from two aged neurons treated with 10 μM retigabine during the time illustrated by the light gray box. No holding current was applied. D. Left: Summary of the resting membrane potential measured before (light purple) and after (dark purple) the application of retigabine. Right: Summary of the hyperpolarization produced by retigabine calculated by subtracting the post-drug voltage from the pre-drug voltage (ΔV). Data points are from $N = 2$ animals, $n = 7$ cells, 120-week-old mice. P-values are shown at the top of the graphs.

